Project: Worked examples
Title: Containment fill, bend radius and cable pulling
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

The fill of a conduit, trunking or tray by the cables in it against the space factor you enter, the bend radius of the route against the multiple of the largest cable diameter you enter, and the pull of the cables through a route of straight lengths and bends in both directions: the tension along the route, the sidewall pressure at each bend, and the pull direction that passes every check (the lower maximum tension when both do or neither does). Space factor, bend multiple, friction coefficient and pulling limits are your own records; no cable or conduit factor method is offered.

WHAT IT CALCULATES	RELATION
Cable area	$A_{cab} = \Sigma\, n_i \cdot \pi \cdot d_i^2/4$ (n_i cables of overall diameter d_i)
Containment area	$A_{int} = \pi \cdot D^2/4$ (conduit, internal diameter D) or $W \cdot H$ (trunking or tray, internal width × depth)
Containment fill	fill = A_{cab}/A_{int}; utilisation u_{fill} = fill/f_{max}, f_{max} the space factor from your record
Bend radius	$R_{min} = n_{bend} \cdot d_{max}$, d_{max} the largest cable diameter, n_{bend} from your record; each bend $u_b = R_{min}/R$
Cable weight	$w = \Sigma\, n_i \cdot m_i$ (kg/m); force per metre $w \cdot g \cdot w_c$, g = 9.80665 m/s², w_c your weight-correction factor (1 = full weight)
Straight length	$T_2 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
Tension round a bend in the horizontal plane, the cable resting on the containment floor (the default)	$T_2 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$, $\mu_e = w_c \cdot \mu$, θ in radians, R the bend radius: the cable's weight bears on the floor round the bend as well as the tension, so a heavy cable on a large radius gains tension even from $T_1 = 0$
Tension round a bend, capstan option (no weight term)	$T_2 = T_1 \cdot e^{(w_c \cdot \mu \cdot \theta)}$, the equation for a weightless pull; it is $T_2 = T_1 \cdot e^{(\mu_e \cdot \theta)}$ and equals the weighted form when $w \cdot g \cdot w_c \cdot R$ is small against T_1
Sidewall pressure at a bend	$SWP = T_2/R$ (N/m)
Pulling direction	the route is run in the listed order and in reverse; the direction that passes every tension and sidewall pressure check is recommended, and when both or neither pass, the one of lower maximum tension (equal tension: lower SWP)
Cable limit	load shared equally, so each cable carries T/N (N cables); $T_{cab} = k_n \cdot N \cdot \min(T_{max,i})$, k_n your derating for more than 3 cables (1 if none); $u_T = \max_i (T/N)/(k_n \cdot T_{max,i}) = T/T_{cab}$
Grip limit	$u_{grip} = T/T_{grip}$, T_{grip} the limit of the grip or pulling eye (your record); sidewall pressure $u_P = SWP_{max}/P_{lim}$

Basis: Other mechanical stresses on wiring systems (damage during installation, bending radius)[1](reg. 522.8); Conduit systems for cable management: general requirements[2](general); Friction: belt friction (the capstan equation)[3](Ch. 8).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
cables	Cables in the containment and pulled together	2 row(s)		spec: Cable data sheets CDS-21 (servo power) and CDS-22 (encoder): overall diameter, mass and maximum pulling tension	any
$cont_{type}$	Containment	Conduit (round)		drawing: CND-ROUTE-07 rev C	any
D_c	Conduit internal diameter (conduit)	40	mm	drawing: CND-ROUTE-07 rev C	10 … 300 mm
f_{max}	Space factor limit (fill limit, from your record)	0.5		reference: record: space factor for this arrangement (a value invented for this example; enter your own from your installation standard or company rule)	0.01 … 1
n_{bend}	Minimum bend radius as a multiple of the largest cable diameter (from your record)	10		reference: record: minimum bend radius multiple (a value invented for this example; enter the largest multiple from the cable data sheets)	1 … 100
route	Pull route in pull order (straight lengths and bends)	6 row(s)		drawing: CND-ROUTE-07 rev C	any
T_1	Tension at the start of the pull (pulling rope, drum or head)	150	N	spec: Pulling method statement PMS-4: 150 N rope and pulling-head tension at the start of the pull	0 … 10000 N
$bend_{model}$	Tension round a bend: horizontal bend with the cable's weight on the containment floor (default), or the capstan equation without the weight term	Horizontal bend, cable on the containment floor (weight term)		default	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
μ	Coefficient of friction, cable sheath on containment (from your record)	0.3		reference: record: coefficient of friction, sheath on lubricated conduit (a value invented for this example; enter your own)	0 … 2
P_{lim}	Sidewall pressure limit, force per metre of bend (from the cable maker)	3000	N/m	reference: record: sidewall pressure limit (a value invented for this example; enter the cable maker's figure)	1 … 10000000 N/m

TABLE ROW	VALUES
cables[1]	name = Servo power; od = 14 mm; count = 2; mass = 0.35 kg/m; T_max = 600 N
cables[2]	name = Encoder and control; od = 9 mm; count = 4; mass = 0.12 kg/m; T_max = 200 N
route[1]	kind = Straight; L = 6 m
route[2]	kind = Bend; theta = 90 deg; R = 500 mm
route[3]	kind = Straight; L = 14 m
route[4]	kind = Bend; theta = 90 deg; R = 400 mm
route[5]	kind = Straight; L = 9 m
route[6]	kind = Bend; theta = 45 deg; R = 600 mm

3. Assumptions

A1. Cables are circular and counted by overall diameter; there is no packing or cable and conduit factor method (the jam-ratio warning is the only jamming check). The space factor is yours, from your own record. (Fill by geometry; the factor is a user record.)
A2. Every cable is pulled together as one bundle on one pulling head with the load shared equally, so each cable carries T/N and the weakest cable governs: the cable limit is N times the lowest maximum pulling tension, optionally derated by your factor for more than 3 cables. The grip or pulling-eye limit is a separate limit you enter. (Equal sharing; unequal sharing is not modelled.)
A3. Straight lengths are horizontal and the cable weight bears on the containment over the full length; bends are taken in the plane of the route. Inclines, vertical runs and pulley or roller reductions are not modelled; fold them into μ. A cradled or triangular weight correction is entered as w_c, which scales the weight on the straight lengths and the friction in the bends (the contact force of a cradled or triangular arrangement is w_c times that of a flat one), so $T_2 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$ and, at a bend, $\mu_e = w_c \cdot \mu$ in the cosh/sinh (or capstan) form. A bend in a vertical plane has the weight along the cable rather than on the floor: take the capstan option and fold the rise into the route. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A4. The coefficient of friction is one constant for the whole route; static and running friction are not distinguished. (A user record with its source.)
A5. Sidewall pressure is the tension leaving the bend over the bend radius, for the bundle as a whole. Enter the cable maker's per-cable limit unchanged: the bundle T/R against it is conservative for a triangular arrangement and roughly equal for a cradled one. Do not multiply the limit by the number of cables. (SWP = T_2/R.)

A6. The bend radius uses the individual cable diameter; for a triplexed or bundled assembly, enter the assembly diameter from the cable maker as the overall diameter. (Cable data sheet.)
A7. The tension at the start of the pull is the same in both directions. With the horizontal-bend (weight) model a heavy cable gains tension at a bend even from $T_1 = 0$; with the capstan option a bend at the very start of a pull with no start tension adds nothing, so enter the pulling rope or drum tension when it matters. (Weight term for a horizontal bend; capstan equation as the option.)

4. Working

[1] $A_{c1} = n \cdot \pi \cdot d^2/4$
$= 2 \times \pi \times (14\ \text{mm})^2 / 4$
$= 307.88\ \text{mm}^2$
Servo power: cross-section of 2 cable(s)

[2] $A_{c2} = n \cdot \pi \cdot d^2/4$
$= 4 \times \pi \times (9\ \text{mm})^2 / 4$
$= 254.47\ \text{mm}^2$
Encoder and control: cross-section of 4 cable(s)

[3] $A_{cab} = \Sigma\ n_i \cdot \pi \cdot d_i^2/4$
$= A_{c1} + A_{c2}$
$= 562.35\ \text{mm}^2$
Total cable cross-section

[4] $OD_{max} = \max\ d_i$
$= 14\ \text{mm}$
$= 14\ \text{mm}$
Largest cable overall diameter

[5] $w = \Sigma\ n_i \cdot m_i$
$= 2 \times 0.35\ \text{kg/m} + 4 \times 0.12\ \text{kg/m}$
$= 1.18\ \text{kg/m}$
Mass per metre of the bundle

[8] $A_{int} = pi * D_c\text{^}2 / 4$
$= \pi \times (40\ \text{mm})\text{^}2 / 4$
$= 1256.6\ \text{mm}^2$
Internal cross-section of the conduit

[10] $fill = A_{cab} / A_{int}$
$= 562.35\ \text{mm}^2 / 1256.6\ \text{mm}^2$
$= 44.75\ \%$
Fill: cable cross-section over internal cross-section

[11] $u_{fill} = A_{cab} / (A_{int} * f_{max})$
$= 562.35\ \text{mm}^2 / (1256.6\ \text{mm}^2 \times 0.5)$
$= 0.895$
Fill utilisation against the space factor

[13] $R_{min} = n_{bend} * OD_{max}$
$= 10 \times 14\ \text{mm}$
$= 140\ \text{mm}$
Minimum bend radius for the largest cable

[14] $u_{b2} = R_{min}/R$
$= 140\ \text{mm} / 500\ \text{mm}$
$= 0.28$
Segment 2: bend radius utilisation

[15] $u_{b4} = R_{min}/R$
$= 140\ \text{mm} / 400\ \text{mm}$
$= 0.35$
Segment 4: bend radius utilisation

[16] $u_{b6} = R_{min}/R$
= 140 mm / 600 mm
= 0.23333
Segment 6: bend radius utilisation
[17] $u_{bend} = \max R_{min}/R$
= 3 bend(s)
= 0.35
Bend radius utilisation (tightest bend)
[20] wg = w * 9.80665 m/s^2
= 1.18 kg/m × 9.80665 m/s^2
= 11.572 N/m
Weight per metre of the bundle on the containment
[22] $Ta_1 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 150 N + 11.572 N/m × 0.3 × 6 m
= 170.83 N
Segment 1 (as listed): tension after the straight length
[23] $Ta_2 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 170.83 N × cosh(0.47124) + √(170.83² + (11.572 N/m × 0.5 m)²) × sinh(0.47124)
= 273.71 N
Segment 2 (as listed): tension after the bend
[24] $Pa_2 = T_2/R$
= 273.71 N / 0.5 m
= 547.43 N/m
Segment 2 (as listed): sidewall pressure
[25] $Ta_3 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 273.71 N + 11.572 N/m × 0.3 × 14 m
= 322.31 N
Segment 3 (as listed): tension after the straight length
[26] $Ta_4 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 322.31 N × cosh(0.47124) + √(322.31² + (11.572 N/m × 0.4 m)²) × sinh(0.47124)
= 516.36 N
Segment 4 (as listed): tension after the bend
[27] $Pa_4 = T_2/R$
= 516.36 N / 0.4 m
= 1290.9 N/m
Segment 4 (as listed): sidewall pressure
[28] $Ta_5 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 516.36 N + 11.572 N/m × 0.3 × 9 m
= 547.6 N
Segment 5 (as listed): tension after the straight length
[29] $Ta_6 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 547.6 N × cosh(0.23562) + √(547.6² + (11.572 N/m × 0.6 m)²) × sinh(0.23562)
= 693.1 N
Segment 6 (as listed): tension after the bend
[30] $Pa_6 = T_2/R$
= 693.1 N / 0.6 m
= 1155.2 N/m
Segment 6 (as listed): sidewall pressure
[31] $Tb_6 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 150 N × cosh(0.23562) + √(150² + (11.572 N/m × 0.6 m)²) × sinh(0.23562)
= 189.89 N
Segment 6 (reversed): tension after the bend

[32] $Pb_6 = T_2/R$
= 189.89 N / 0.6 m
= 316.49 N/m
Segment 6 (reversed): sidewall pressure
[33] $Tb_5 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 189.89 N + 11.572 N/m × 0.3 × 9 m
= 221.14 N
Segment 5 (reversed): tension after the straight length
[34] $Tb_4 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 221.14 N × cosh(0.47124) + √(221.14² + (11.572 N/m × 0.4 m)²) × sinh(0.47124)
= 354.28 N
Segment 4 (reversed): tension after the bend
[35] $Pb_4 = T_2/R$
= 354.28 N / 0.4 m
= 885.7 N/m
Segment 4 (reversed): sidewall pressure
[36] $Tb_3 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 354.28 N + 11.572 N/m × 0.3 × 14 m
= 402.88 N
Segment 3 (reversed): tension after the straight length
[37] $Tb_2 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 402.88 N × cosh(0.47124) + √(402.88² + (11.572 N/m × 0.5 m)²) × sinh(0.47124)
= 645.43 N
Segment 2 (reversed): tension after the bend
[38] $Pb_2 = T_2/R$
= 645.43 N / 0.5 m
= 1290.9 N/m
Segment 2 (reversed): sidewall pressure
[39] $Tb_1 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 645.43 N + 11.572 N/m × 0.3 × 6 m
= 666.26 N
Segment 1 (reversed): tension after the straight length
[40] $T_{max,a}$ = T at the end of the route, as listed
= 693.1 N
= 693.1 N
Maximum tension, pulling in the listed order
[41] $T_{max,b}$ = T at the end of the route, reversed
= 666.26 N
= 666.26 N
Maximum tension, pulling in the reverse order
[42] $P_{max,a}$ = max SWP, as listed
= 1290.9 N/m
= 1290.9 N/m
Maximum sidewall pressure, pulling in the listed order
[43] $P_{max,b}$ = max SWP, reversed
= 1290.9 N/m
= 1290.9 N/m
Maximum sidewall pressure, pulling in the reverse order
[44] $T_{cab} = N \cdot \min(T_{max,i})$
= 6 × 200 N
= 1200 N
Pulling tension limit of the bundle from the cables (equal load share, weakest cable)

[46] $max_{tension}$ = min($T_{max,a}$; $T_{max,b}$)
= 693.1 N ; 666.26 N
= 666.26 N
Maximum pulling tension, recommended direction
[47] P_{max} = max SWP in the recommended direction
= 1290.9 N/m
= 1290.9 N/m
Maximum sidewall pressure, recommended direction
[48] u_{Tc1} = (T/N)/($k_n \cdot T_{max,i}$)
= 111.04 N / (1 × 600 N)
= 0.18507
Servo power: share of the pull against its own limit
[49] u_{Tc2} = (T/N)/($k_n \cdot T_{max,i}$)
= 111.04 N / (1 × 200 N)
= 0.55521
Encoder and control: share of the pull against its own limit
[50] u_T = $max_{tension}$ / T_{cab}
= 666.26 N / 1200 N
= 0.55521
Tension utilisation against the cable limit (the weakest cable)
[51] u_P = P_{max} / P_{lim}
= 1290.9 N/m / 3000 N/m
= 0.43028
Sidewall pressure utilisation against the limit

5. Results

SYMBOL	VALUE	DESCRIPTION
A_{cab}	562.35 mm²	Total cable cross-section
A_{int}	1256.6 mm²	Internal cross-section
fill	44.75 %	Fill
u_{fill}	0.895	Fill utilisation
OD_{max}	14 mm	Largest cable diameter
w	1.18 kg/m	Mass per metre of the bundle
R_{min}	140 mm	Minimum bend radius
u_{bend}	0.35	Bend radius utilisation
$T_{max,a}$	693.1 N	Maximum tension, as listed
$T_{max,b}$	666.26 N	Maximum tension, reversed
$P_{max,a}$	1290.9 N/m	Maximum sidewall pressure, as listed
$P_{max,b}$	1290.9 N/m	Maximum sidewall pressure, reversed
$max_{tension}$	666.26 N	Maximum tension, recommended direction
P_{max}	1290.9 N/m	Maximum sidewall pressure, recommended direction
T_{cab}	1200 N	Pulling tension limit from the cables
u_T	0.55521	Cable tension utilisation
u_P	0.43028	Sidewall pressure utilisation

SYMBOL	VALUE	DESCRIPTION
direction	"Reverse: pull from segment 6 to segment 1"	Recommended pull direction

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fill against the space factor	u_{fill} = 0.895	u_{fill} ≤ 1	0.895	
PASS	Pulling tension against the weakest cable's limit	u_T = 0.55521	u_T ≤ 1	0.555	
PASS	Sidewall pressure against the limit	u_P = 0.43028	u_P ≤ 1	0.43	
PASS	Bend radius against the cable minimum	u_{bend} = 0.35	u_{bend} ≤ 1	0.35	[1](reg. 522.8)
PASS	Largest cable fits the containment	largest cable 14 mm	conduit internal diameter 40 mm	0	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: f_{max} (Containment space factor); n_{bend} (Minimum bend radius multiple); μ (Pulling friction coefficient); P_{lim} (Pulling tension, grip and sidewall pressure limits). Enter each as one so its source, edition and licence travel with it.
WARNING — input.bend_model: $bend_{model}$ (Tension round a bend: horizontal bend with the cable's weight on the containment floor (default), or the capstan equation without the weight term) was not entered; the template default 'Horizontal bend, cable on the containment floor (weight term)' is used.

8. Verification

Template elec.containment@1.2.0 Content hash 9c593a68e2b7eb0d Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. BS EN 61386-1.
3. Beer and Johnston, Vector Mechanics for Engineers: Statics.